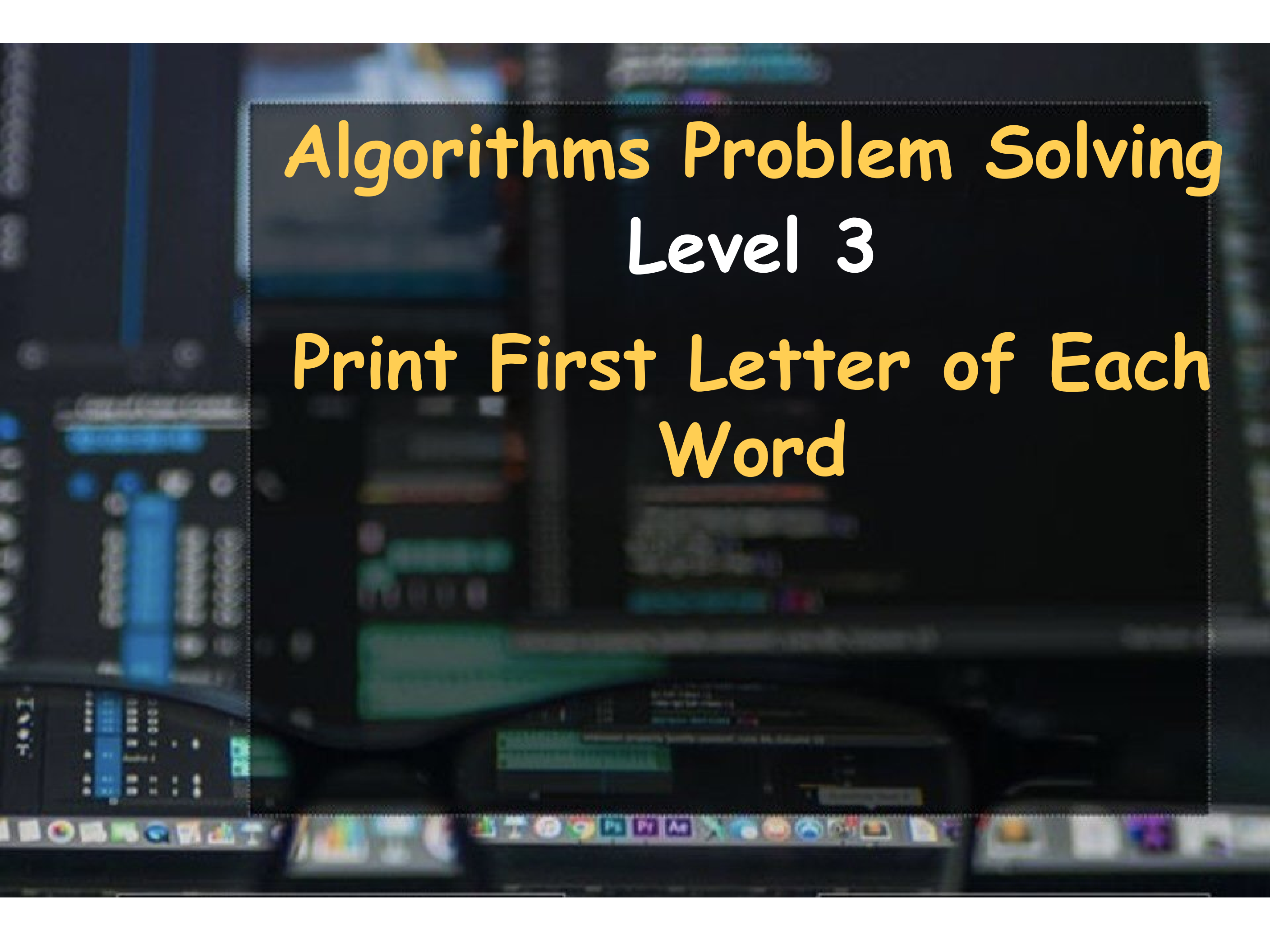
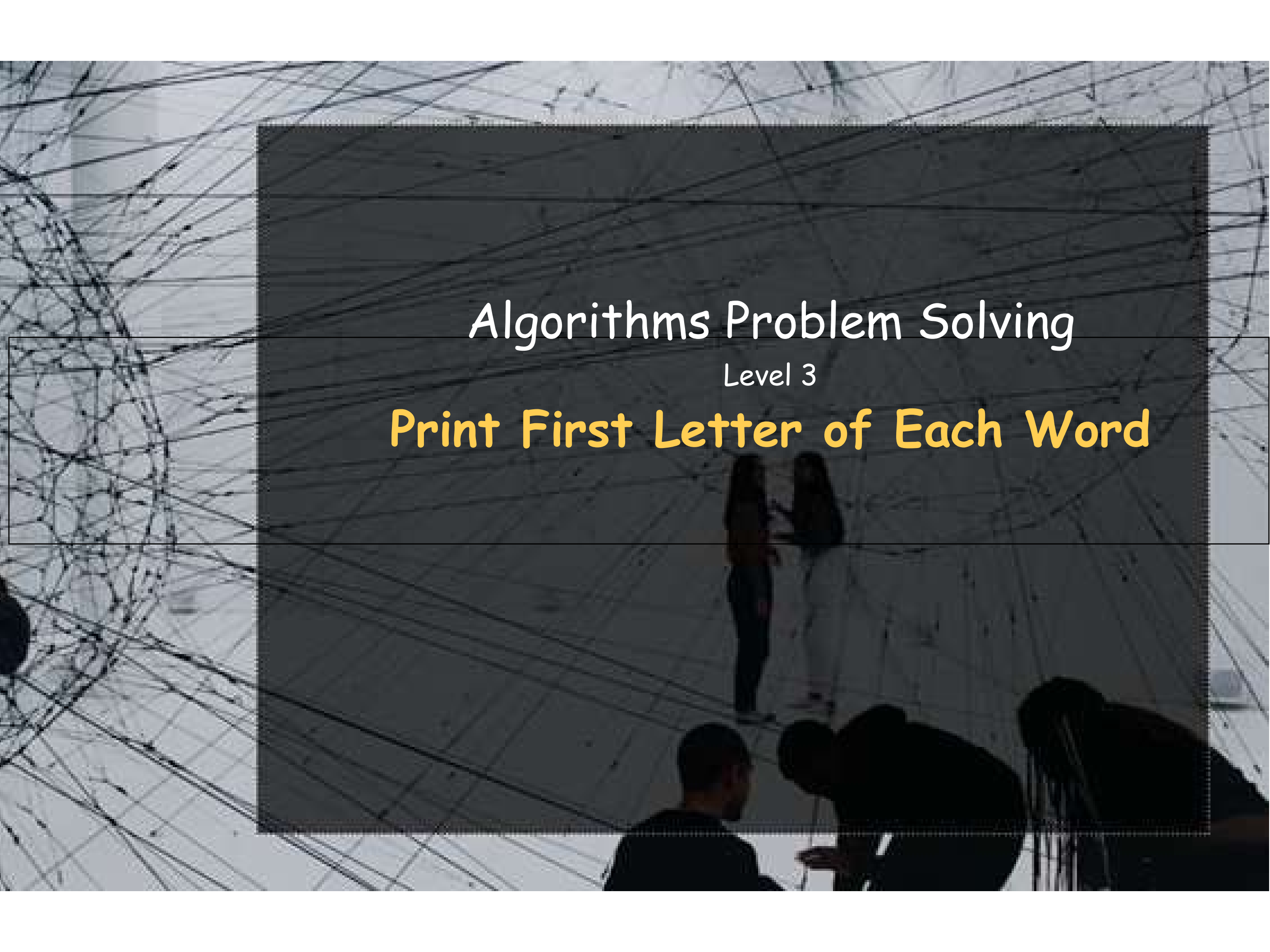


The background of the slide is a blurred image of a computer monitor. On the left side of the monitor, a calculator application is visible, showing a numeric keypad and some text. The main area of the monitor displays lines of code in a dark-themed editor, with some text highlighted in blue and green. At the bottom of the monitor, a taskbar is visible with various application icons.

Algorithms Problem Solving Level 3

Print First Letter of Each Word



Algorithms Problem Solving

Level 3

Print First Letter of Each Word

Definition

Initial Letter Extraction

 Program reads a complete sentence and extracts the first letter of every word.

 Word starts are detected by identifying the beginning of the string or a position following whitespace.

Explanation

Full String Input

-  ``getline()`` is used so that the complete sentence, including spaces, can be processed.

Word Boundary Detection

-  A character is considered the beginning of a word when it is at index ``0`` or follows a space.

-  A more robust implementation can use ``isspace()`` to recognize different whitespace characters.

Character Extraction

-  When a word boundary is detected, the current character is printed.

-  The original character casing can be preserved.

Procedures

1. Read the complete sentence using ``getline()'`.
2. Traverse the string character by character.
3. Detect the beginning of each word.
4. Print the first character of each detected word.

💡 Example

Input:

Sara Alterkawi

➡ Result

S

A

Note

- ◆ A state-based approach can handle multiple consecutive spaces more reliably.
- ◆ The method can be extended to create acronyms.
- ◆ Word-boundary logic can be adapted for tabs and other whitespace characters.

⊘ Warning

- ◆ ``cin >> str`` reads only the first word, so it is unsuitable for complete sentences.
- ◆ Directly accessing ``str[0]`` requires checking that the string is not empty.
- ◆ Multiple spaces can produce incorrect detections if every space is treated as a guaranteed word boundary.



Comparison

▨ State-Based Parsing

- ✓ Tracks whether the program is currently inside a word.
- ✓ Handles multiple consecutive whitespace characters correctly.
- ✓ Avoids creating temporary word strings.

▤ Split-Based Parsing

- ✓ Splits the sentence into separate word strings.
- ✓ Takes the first character from every token.
- ✓ Requires additional vector storage.

✦✦ Summary

- ✓ Reads complete text input.
- ✓ Detects word boundaries during character traversal.
- ✓ Extracts the leading character of each word.
- ✓ Demonstrates a foundation for acronym and token-processing algorithms.

Thank you 😊